



JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR

Website: www.jdcem.ac.in E-mail: info@jdcem.ac.in

(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai

Department of Computer Science And Engineering

"A place to learn; A chance to grow"

2025-2026(Odd Semester)



MSE-2 EXAMINATION

Course: B Tech in Computer Science and Engineering

Year/Semester: 5th Semester (3rd Year)

Subject: Elective -V (React JS)

Subject Code: CS5E002C

Max Marks: 20

Date: 04/10/25

Duration: - 1 Hr.

Instructions to the Students:

1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

Students should be able to:

- CO1:** Understand the fundamentals of React and JSX.
CO2: Design and develop functional UI components using hooks and props.
CO3: Manage application state using use State, use Effect, and Context API.
CO4: Integrate APIs and handle routing in React applications.
CO5: Build, deploy, and optimize a full-scale single-page React application.

SECTION A

1. All Questions are compulsory.

2. Answer in one word/sentence.

Q. No.	Questions	Level /CO Mapped	Marks
Q.1	State the meaning of Route Params in React Router v6.	L1/CO1	1
Q.2	Identify the purpose served by the useContext () hook in React applications.	L2/CO1	1
Q.3	List two applications of LocalStorage in React development.	L1/CO1	1
Q.4	Give the benefit of Axios over the fetch API through a short example.	L2/CO1	1

SECTION B

1. All Questions are compulsory.

2. Solve answer in brief (two, three Sentence).

Q.5	Differentiate between data passing through Props and data sharing through Context API.	L4/CO1	2
Q.6	Illustrate the concept of Nested Routes in React Router v6 with an example.	L3/CO1	2
Q.7	Discuss the process of maintaining loading states while integrating APIs in React.	L2/CO1	2
Q.8	Apply the concept of LocalStorage to store and retrieve a user's theme preference (dark / light).	L3/CO2	2

SECTION C

1. Solve any two questions.

Q.9	Analyze the methods of component communication through Props, Context API, and Custom Hooks, and conclude the most suitable approach for large-scale applications.	L4/CO2	4
Q.10	Evaluate the design of a product listing feature in a React application with reference to Search, Filters, and Pagination, and justify the approach that enhances user experience.	L6/CO2	4
Q.11	Create a functional React component to integrate API data using Axios, implement loading and error handling states, and display the output in a structured list format.	L5/CO2	4